~~7.5~~ 8.0 Aeyan

EGC 211P Class Quiz I

Time: 30 minutes Maximum Marks: 25
Roll Number: BC2025051 Name: Kushman Goyal

Use both sides of the page.

(1) **Q1.** [1 mark] In C++, what is the difference between a `method` and a `function`?

A function could be defined anywhere however methods are functions initialized inside classes.

(2) **Q2.** [2 marks] What is the difference between enapsulation and abstraction?

~~Encaps~~

Abstraction is the idea of hiding much of the technical intricacies from the user while Encapsulation is the process of applying this idea using code.

ex- Abstraction means that while a cat is a cute animal for a kid, it is a complex collection of biological processes for a veterenarian. Encapsulation is how the biological processes are encapsulated to make it feel like cute animal to the user i.e the kid.

Q3. [2 marks] One `struct` has 9 attributes (assume of size a bytes each), and has 3 member functions (assume each occupies m bytes of characters). What will be the size of any object of that `struct`, and why?

(1) + (0.5)

The size of any object of that struct will be '9a' upon initialization because the 9 attributes of 'a' bytes each characterize the object and must be assigned values for initializing the struct while the 3 member functions may or may not be called

Q4. [2 marks] Book has pages. What are the entities in this statement, and what kind of relationship holds between them? Explain.

(0.5)

In the statement 'Book has pages', the entities are 'Books' and 'Pages'. The two entities are related by Aggregration because a Book 'has' pages, which still have their independat existence with or with a book, just like wheels to a car.

Q5. [6x3 = 18 marks] In YouTube (the real software), users can comment on videos. Let user *ahana* follow user *stroustrup*, and put comment c on *stroustrup*'s video v about how to frame easy looking C++ questions that make the clock run faster than usual. List down all the relationships that you can identify between every pair (${}^4C_2$ combinations) in the four entities: *ahana*, *stroustrup*, c, and v. For each, give your reason.

P. T. O →